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CSE 112 Computer Organization: Quiz 3 (Set A)

Maximum Marks: 35

Q1. For the below set of instructions, find all the Read-after-Write (RAW) hazards for a processor with five stages: fetch(F), decode (D), execute (EX), memory access (MA), and register writeback (RW). Mention the instruction number and the registers that are causing the hazard. Answer in the format given below: **[6 Marks]**

Register	Instruction number that writes to the specified register	Instruction number that reads from the specified register

For any instruction, the syntax is:

INST Dest.Register, Src.Register1, Src.Register2

OR

INST Dest.Register, Src.Register, #ImmValue

1. subi R6, R7, #2
2. add R5, R6, R2
3. mul R9, R11, R5
4. mul R8, R10, R9
5. sub R1, R8, R4
6. add R8, R1, R3

Q2. Consider processor A, which executes a program without a pipeline, and each instruction takes one clock cycle to execute. Consider processor B, which executes a program with a **4-stage** pipeline implementation with stages S1, S2, S3, and S4. Each pipeline stage is allocated one clock cycle to complete. For the given program, answer the following questions. Ignore data hazards (if any). **[16 Marks]**

1 add r1, r2, r3
2 sub r4, r5, r6
3 add r2, r15, r3
4 add r6, r3, r9
5 sub r5, r7, r8
6 add r15, r11,r13
7 add r10, r11, r12
8 sub r14, r11, r13

a. If the clock duration for processor A is 25 μs, determine the time required to implement the program in processor A. **[1 Mark]**
b. For processor B, if stage S1 takes 5 μs, S2 takes 3 μs, S3 takes 10 μs, and S4 takes 6 μs respectively, determine the clock duration of the processor. Assume that pipeline registers do not add any delay to the path. **[1 Mark]**
c. Draw the pipeline diagram when processor B implements the above program. **[12 Marks]**
d. Determine the time required to implement the program in processor B. **[1 Mark]**

e. What is the speed-up obtained when processor A is replaced by processor B? (Hint: Take the ratio of time required to implement the program in processor A and processor B) **[1 Mark]**

Q3. Imagine a virtual ISA having an instruction size of 32 bits and 16 registers. Suppose the ISA supports an address space of 1 MegaByte with byte-addressable memory. The number of unique Instructions/Operations supported by the processor is 32. **[7 Marks]**

Syntax for Memory type instruction : <Filler Bits><Opcode><Memory address> <Register address> Syntax for Register type instruction : <Opcode><Source Register Address> <Destination Register Address> <Filler Bits>

Note: In regards to the above syntax, the left side is the most significant bit and the right side is least significant bit.

a. What are the minimum bits required to represent the opcode of instruction? **[1 Mark]**
b. What are the minimum bits required to represent a register uniquely? **[1 Mark]**
c. What are the minimum bits required to represent a memory address uniquely? **[1 Mark]**
d. How many filler bits are required in memory type and register type instruction respectively? **[1Mark]**
e. For a memory type instruction with hexadecimal value 100001AB, determine the value of opcode, memory address, and register address. **[1*3 = 3 Marks]**

Q4. Consider a stack that uses the 'PUSH' and 'POP' instructions (defined below). For the given program, write the value of the register modified after each 'POP' instruction. Initially, the value of R1 is 5, R2 is 6, R4 is 7, R9 is 11 and R7 is 21. **[6 Marks]**

Push	Pushes the data stored in reg1 onto the stack	PUSH reg1
Pop	Pops the data stored on the top of the stack into reg1	POP reg1

PUSH R1
PUSH R2
POP R3
PUSH R4
POP R5
POP R4
PUSH R7
PUSH R4
PUSH R9
POP R10
POP R11
POP R12